

Overview of Random forest

Subject: Random forest

1.

which is equal to the mean of the Y_i 's falling in the cells containing $\mathbf{x}$ in the forest. If we define the connection function of the M finite forest as $K_{M,n}(\mathbf{x}, \mathbf{z}) = \frac{1}{M} \sum_{j=1}^M 1_{\mathbf{z} \in A_n(\mathbf{x}, \Theta_j)}$, i.e. the proportion of cells shared between $\mathbf{x}$ and $\mathbf{z}$, then almost surely we have $m \sim M, n(\mathbf{x}, \Theta_1, \dots, \Theta_M) = \sum_{i=1}^n Y_i K_{M,n}(\mathbf{x}, \mathbf{x}_i) \xrightarrow{P} \sum_{i=1}^n K_{M,n}(\mathbf{x}, \mathbf{x}_i) = \frac{1}{n} \sum_{i=1}^n K_{M,n}(\mathbf{x}, \mathbf{x}_i)$, which defines the KeRF.

2.

Relation between infinite KeRF and infinite random forest When the number of trees M goes to infinity, then we have infinite random forest and infinite KeRF. Their estimates are close if the number of observations in each cell is bounded:

3.

$|m_{M,n}(\mathbf{x}) - m \sim M, n(\mathbf{x})| \leq b_n - a_n \leq \frac{b_n - a_n}{a_n} m \sim M, n(\mathbf{x})$.

4.

Uniform KeRF Uniform KeRF is built in the same way as uniform forest, except that predictions are made by $m \sim M, n(\mathbf{x}, \Theta_1, \dots, \Theta_M)$, the corresponding kernel function, or connection function is

5.

The number B of samples (equivalently, of trees) is a free parameter. Typically, a few hundred to several thousand trees are used, depending on the size and nature of the training set. B can be optimized using cross-validation, or by observing the out-of-bag error: the mean prediction error on each training sample x_i , using only the trees that did not have x_i in their bootstrap sample. The training and test error tend to level off after some number of trees have been fit.

6.

Consistency of uniform KeRF Providing $k \rightarrow \infty$ and $n/2^k \rightarrow \infty$, there exists a constant $C > 0$ such that,

7.

Consistency results Assume that $Y = m(\mathbf{X}) + \varepsilon$, where ε is a centered Gaussian noise, independent of $\mathbf{X}$, with finite variance $\sigma^2 < \infty$. Moreover, $\mathbf{X}$ is uniformly distributed on $[0, 1]^d$ and m is Lipschitz. Scornet proved upper bounds on the rates of consistency for centered KeRF and uniform KeRF.

8.

$$\mathbb{E} [\|\tilde{m}_n - m\|_{\mathcal{C}}^2] \leq C \frac{1}{n} \frac{1}{(3 + d \log 2)} (\log n)^2 \leq C_1 n^{-1/(3+d \log 2)} (\log n)^2.$$

9.

Consistency of centered KeRF Providing $k \rightarrow \infty$ and $n/2^k \rightarrow \infty$, there exists a constant $C_1 > 0$ such that, for all n ,

10.

or by taking the plurality vote in the case of classification trees. This bootstrapping procedure leads to better model performance because it decreases the variance of the model, without increasing the bias. This means that while the predictions of a single tree are highly sensitive to noise in its training set, the average of many trees is not, as long as the trees are not correlated. Simply training many trees on a single training set would give strongly correlated trees (or even the same tree many times, if the training algorithm is deterministic); bootstrap sampling is a way of de-correlating the trees by showing them different training sets. Additionally, an estimate of the uncertainty of the prediction can be made as the standard deviation of the predictions from all the individual regression trees on $\mathbf{x}'$:

11.

Preliminaries: Centered forests Centered forest is a simplified model for Breiman's original random forest, which uniformly selects an attribute among all attributes and performs splits at the center of the cell along the pre-chosen attribute. The algorithm stops when a fully binary tree of level k is built, where $k \in \mathbb{N}$ is a parameter of the algorithm.

12.

$$\sigma = \sqrt{\frac{1}{B-1} \sum_{b=1}^B (f_b(\mathbf{x}') - \bar{f})^2}.$$

13.

Variants Instead of decision trees, linear models have been proposed and evaluated as base estimators in random forests, in particular multinomial logistic regression and naive Bayes classifiers. In cases that the relationship between the predictors and the target variable is linear, the base learners may have an equally high accuracy as the ensemble learner.

14.

Using out-of-bag error as an estimate of the generalization error. Measuring variable importance through permutation. The report also offers the first theoretical result for random forests in the form of a bound on the generalization error which depends on the strength of the trees in the forest and their correlation.

15.

Prefiltering: Eliminate features that are mostly just noise. Enriched Random Forest (ERF): Use weighted random sampling instead of simple random sampling at each node of each tree, giving greater weight to features that appear to be more informative. Tree-weighted random forest (TWRF): Give more weight to more accurate trees.